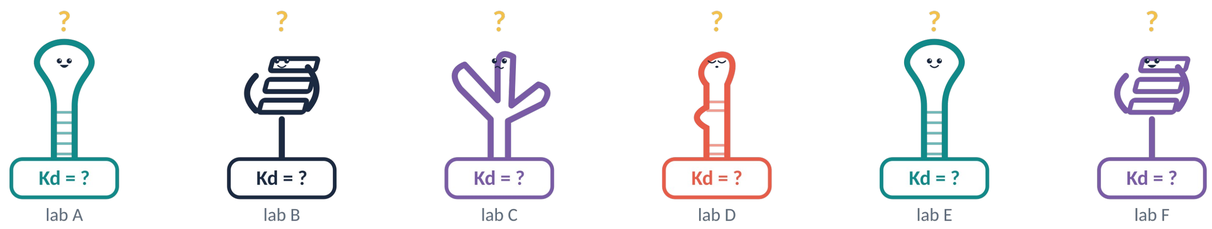
Salt prepares the hand
and weakens the handshake

Why six published norovirus aptamers cannot simply be ranked — and how we plan to compare them fairly

[Your name] · [Lab, Institute] · [email] · PhD Retreat 2026

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1 Six aptamers, six histories



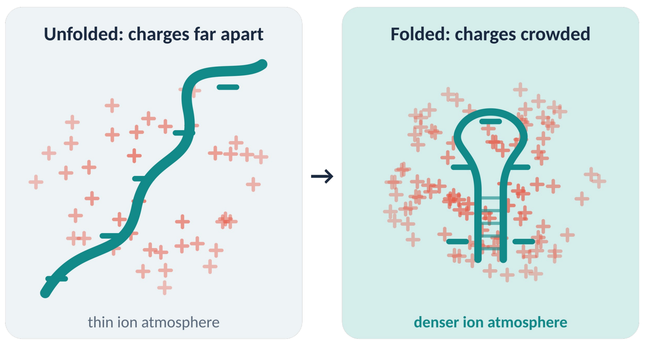
Different labs, techniques, targets and buffers — so the published affinities are not comparable.

An aptamer is not a rigid object with a fixed binding strength: it explores different shapes, and the solution decides how many molecules are in a binding-competent shape. Each one must be optimised in its own conditions.

A salt-dependent change can therefore look like a difference between aptamers when it is really a difference between conditions.

2 Folding costs charge

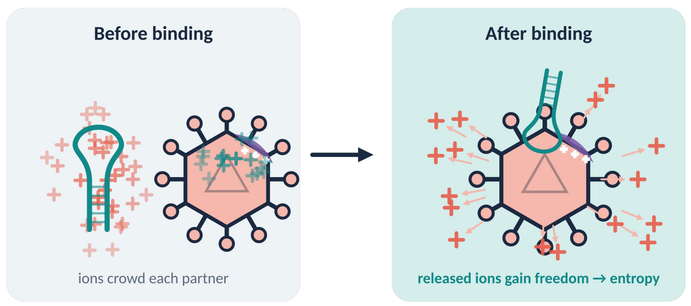
1 · Electrostatics of folding



- Folding pulls negative phosphates together — an electrostatic penalty
- Positive ions screen it: an ion atmosphere — diffuse and mobile, not fixed ions
- Folding and ion uptake are coupled: more salt can stabilise a compact fold

4 Counterion release

3 · Entropy



- Ions are enriched near the DNA and near positive patches on the target
- On association some are released into bulk solution and gain freedom → entropy
- As bulk salt rises there is less to gain from release, so the advantage shrinks
- One reason binding weakens at high salt

5 Identity, not only amount

4 · Which ion, not just how much

K^+



sits in the G-quadruplex channel — a structural role, not just screening

Mg^{2+}



strongly shapes nucleic-acid structure and competes with monovalent ions

Cl^- vs glutamate$^-$



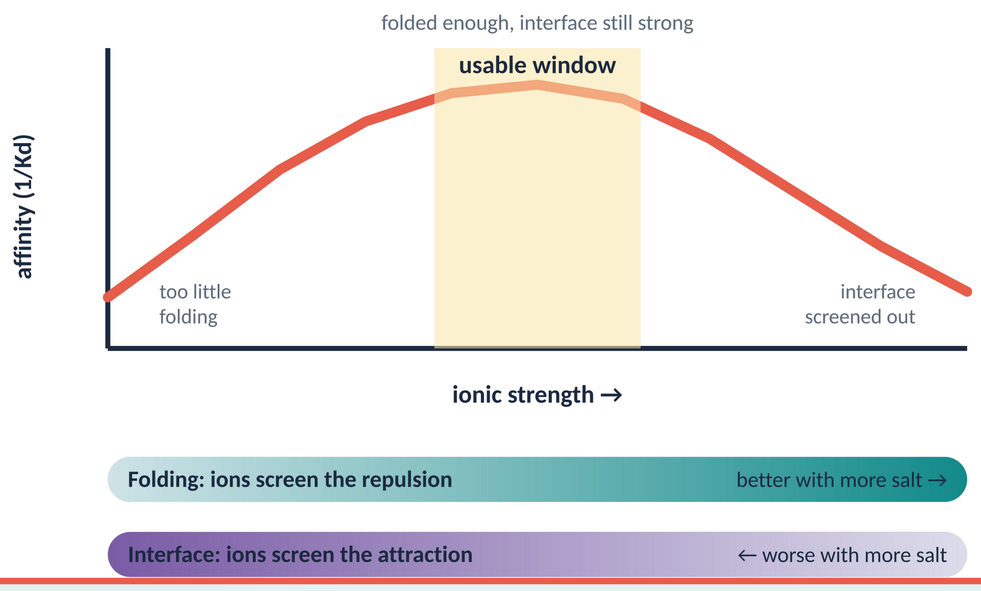
anions are not interchangeable either — swapping them shifts affinity

Two buffers are not equivalent just because both contain "salt".

Ion identity, concentration and pH are separate variables — and we treat them that way.

3 ...and the same ions weaken the handshake

2 · Electrostatics at the interface



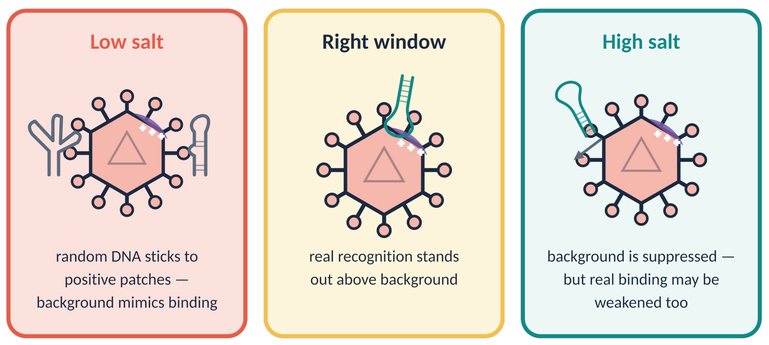
- Binding to a positively charged patch is helped by attraction to the negative backbone
- Salt screens that attraction too — so the ions that help the fold weaken part of the contact
- Affinity against salt is therefore often not monotonic: there can be a window where folding is good and the interface is still strong

What salt dependence does not tell us

A steep fall in affinity with rising salt suggests a charge-driven interaction. It does not tell us how many ions were released: structural change, hydration and other contacts also contribute. We treat the salt response as a clue, not a full explanation.

6 Specificity

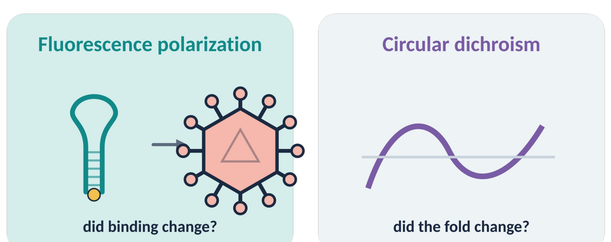
5 · A strong signal is not proof of recognition



Controls, under matched conditions: scrambled sequences · motif mutants · unrelated viral particles · norovirus-negative stool

7 Why fluorescence polarisation

6 · One method, one variable: the solution



- Everything in solution — nothing immobilised, no surface
- No wash: one buffer start to finish, at equilibrium
- One plate scans ion identity, concentration and pH for all six
- With CD: tells a change in fold from a change at the interface

The plan in one line: test all six head-to-head by FP, vary ions and pH deliberately, and compare both how well each aptamer binds and how much its binding changes between conditions.

Takeaway: salt helps the fold, weakens the handshake, and changes the entropy of binding — so the binding we measure belongs to the sequence ***and its solution together***.

Fair buffer: each aptamer's own, or one shared condition for all six?

How far can stool be diluted before the question stops being realistic?

Your experience with FP or CD in complex matrices — what caught you out?